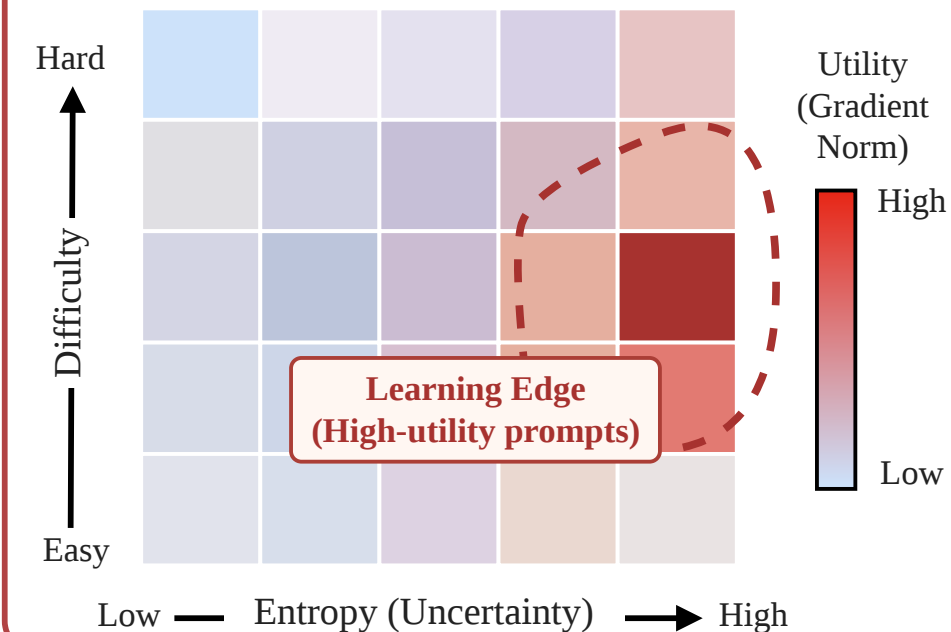


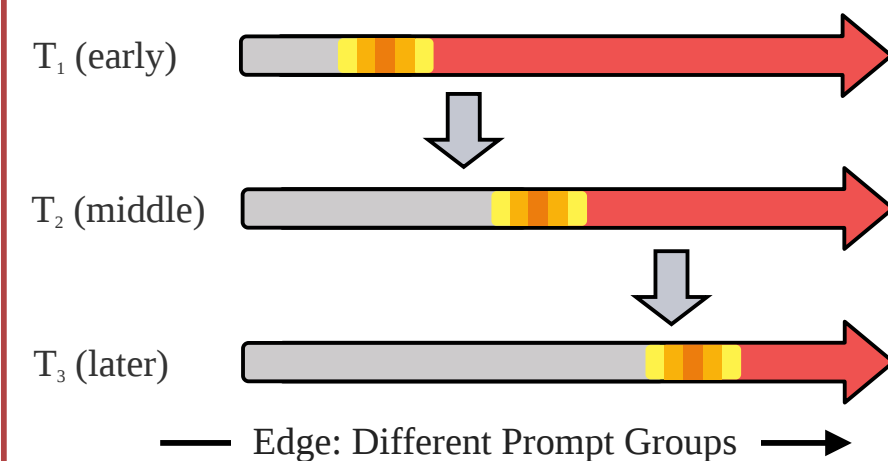
(A) The moving learning edge

A1. Where is the learning edge?

Utility concentrates at intermediate difficulty and high uncertainty.



A2. The edge moves as the policy improves.



Not one-shot ranking, but moving-edge tracking.

(B) Why prior methods fail

Existing methods fail to be both online and efficient in selecting high-utility prompt for RL training.

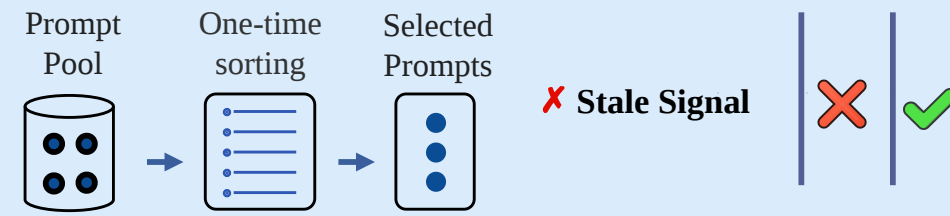


Online

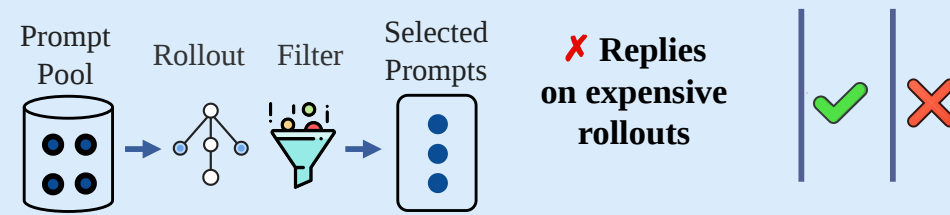


Efficient

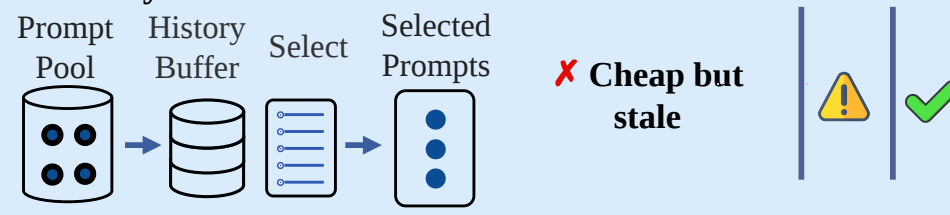
1. Static / offline filters



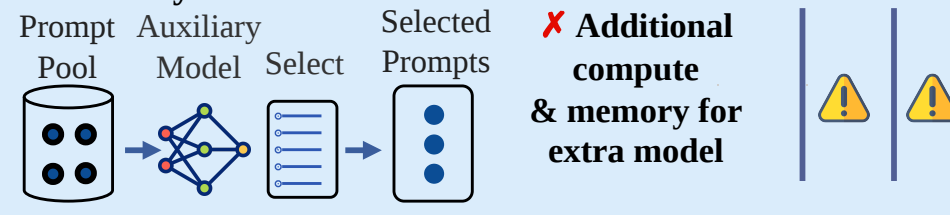
2. Rollout-based online filters



3. History-based estimators



4. Auxiliary-model selectors



(C) HIVE: history-informed narrowing + online verification

